

# 3D Modeling & Design

## 1. Basic Requirements

The following are supplementary requirements for 3D Modeling & Design.

### Applications

- Web browser (Chrome, Edge, or Safari)
- LocalSend
- Onshape FTC parts library plugin — <https://www.onshape.com/> (to be confirmed)

### Online Accounts

- CycleZLab — <https://www.cyclezlab.com/>
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## 2. Environment Setup

The following environment must be configured before you begin. If you have any questions, please consult an administrator.

### Web Browser

Bookmark the following additional websites:

| Name         | URL                                                                     |
|--------------|-------------------------------------------------------------------------|
| goBILDA      | <a href="https://www.gobilda.com/">https://www.gobilda.com/</a>         |
| REV Robotics | <a href="https://www.revrobotics.com/">https://www.revrobotics.com/</a> |
| CycleZLab    | <a href="https://www.cyclezlab.com/">https://www.cyclezlab.com/</a>     |

## 3. Modeling Essentials

### Part Design and Assembly

When performing 3D modeling in Onshape, pay special attention to the following:

- **Dimensional accuracy:** ensure all dimensions in the model match the physical parts
- **Assembly constraints:** set up part-to-part assembly constraints properly
- **Interference check:** run interference checks after assembly
- **Part naming:** name parts and assemblies according to unified conventions

### Common FTC Parts Libraries

Commonly used FTC parts suppliers and libraries:

- **goBILDA:** provides a complete range of FTC structural components
- **REV Robotics:** provides control system modules and structural components
- **CycleZLab:** domestic parts supplier

### Design Standards

- Use metric units (mm)

- Leave appropriate clearance for all 3D-printed parts (0.2–0.3 mm recommended)
- Export formats: STEP files for machining, STL files for 3D printing